



Solar tree



Wi-Fi stand in New Delhi

Solar trees are an economical option for solar implementation in cities. Central Mechanical Engineering Research Institute, Durgapur, West Bengal—a constituent laboratory of Council of Scientific and Industrial Research (CSIR)—has designed and deployed solar trees across villages in Durgapur to mitigate the lack of electricity with renewable energy. In 2017, 2.5kW-capacity solar trees were installed at Sriyani Auditorium, Durgapur to power the gardens and lawns off the grid.

Charkha Park in Connaught Place, New Delhi has also received solar trees for storing solar energy for off-grid use and accommodating citizens to sit under these while charging their cellphones.

Solar rooftops are usual elements of major green commercial buildings across India. Solar-powered streetlights with inbuilt batteries are gaining prom-

inence as highways and remote areas face challenges in reliable grid supply and cabling. For example, highways along Noida-Greater Noida stretch are slowly adopting solar streetlights.

A large volume of solar projects, including some of the world's largest solar parks, is in progress, too. Karnataka, Andhra Pradesh, Uttar Pradesh, Gujarat and Maharashtra are some of the leading Indian states in terms of solar installation. All these initiatives will enable India reduce its dependency on non-renewable fuels and mitigate the nation-wide demand for electricity.

Smart infrastructure for connected cities

Sensors play an important role in shaping smart cities. Urban development and sustainability depend on real-time action, which can be taken based on data shared by such sensors. One of the most popular applications of sensors is in lighting infrastructure, that is, smart lighting.

For example, the city of Bhopal in Madhya Pradesh has installed 400 smart poles and 20,000 LED streetlights across the city. Replacing existing fluorescent streetlights with LEDs was one of the first steps that India took towards energy efficiency.

Smart poles installed in Bhopal deliver remote- and auto-control of streetlights, employ mobile-based SoS notifications to control centres, generate Wi-Fi hotspots and come with integrated surveillance cameras and air quality monitors. Bhopal Smart City authority shares that this infrastructure will lead to zero operating expenditure and maintenance for the next 15 years—saving a substantial amount of money and energy through optimal usage control.

Another example of smart application is the real-time monitoring of water

bodies. Jamshedpur, with the help of Jamshedpur Utilities and Services Companies (JUSCO), has installed sensors in lakes to monitor water quality and aquatic ecology consistently. Any changes in the aquatic composition enable it to take quick action.

Bengaluru in Karnataka is using a smart control platform to reduce traffic congestions. The Electronic City township in Bengaluru is currently running pilots with technology powered by surveillance cameras with computer vision, sensors and an artificial intelligence (AI) engine to assess traffic conditions and recommend rerouting suggestions in real time. The platform is also expected to automate traffic signal controls.

Many other Indian cities have launched real-time parking management platforms that help in reducing on-road vehicle congestion. As a result, gases and particles emitted by vehicles—one of the most dangerous sources of urban pollution—are reduced significantly.

Smart grids and smart meters are available across the country to allow utilities and consumers check the real-time supply and usage status of resources like electricity and water. This reduces resource wastage and promotes consistent distribution across an area.

All these are possible through well-distributed networks across cities. Each infrastructure is Wi-Fi-enabled or uses cellular networks to run. For instance, New Delhi has multiple Wi-Fi booths by Signpost across various parts of central Delhi that deliver open, public Wi-Fi services to its citizens.

Jamshedpur is covered all over by LoRaWAN network, which powers all its public services. There are more such examples.

We are just a few years down the line since the beginning of the ambitious Smart Cities project. While a lot still remains to be explored and done, the progress is quite significant. Not only utilising the newer technologies being installed, but also safeguarding these infrastructures as responsible citizens is crucial for our progress. Because, smart cities are the precursor to sustainable green cities. **EFY**

